A strict strip for Ehrhart roots in dimensions six and seven

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Abstract

We prove that every root z of the Ehrhart polynomial of a reflexive lattice polytope of dimension six or seven satisfies $-3 < \operatorname{Re} z < 2$. This proves the nonreal-root assertion in MathDB 330132, namely Conjecture 6.4 of the 2015 Hegedűs–Higashitani–Kasprzyk manuscript. The proof applies to a larger cone of palindromic binomial-basis polynomials. Its only computation is an exact determinant expansion: all 112 coefficients used in the proof are printed in the appendix and verified by an independent integer-arithmetic program.

Task 1: the original problem and the stronger result

For a six- or seven-dimensional reflexive polytope P, the question asks whether a root $-1/2+u+iv$ of L_P, with $u, v \neq 0$, necessarily has $|u| < 5/2$. This is Conjecture 6.4 on page 15 of [1]. The final article [2] omits the preprint's conjecture section; its page 5 still explicitly leaves the six- and seven-dimensional strip case unproved. We address the preprint's complete assertion, with the stronger conclusion that also includes real roots.

Theorem 1. *Let $d \in \{6, 7\}$ and*

$$F(z) = \sum_{j=0}^{d} h_j \binom{z+d-j}{d}, \qquad h_j = h_{d-j},$$

where

$$h_0 > 0, \quad h_1 \geq h_0, \quad h_2 \geq h_1, \quad h_3 \geq h_1.$$

Then every root of F satisfies $-3 < \operatorname{Re} z < 2$. In particular, every root of the Ehrhart polynomial of a reflexive lattice polytope in either dimension satisfies this strict strip bound.

The last assertion follows from the standard Ehrhart expansion and Hibi's lower bound: the h^*-vector of a reflexive polytope is palindromic, $h_0 = h_d = 1$, $h_1 \geq 1$, and $h_j \geq h_1$ for $2 \leq j \leq d-1$. These facts and their original references are stated on page 2 of [2]. No unimodality assumption is made.

Write

$$h_0 = a, \quad h_1 = a+b, \quad h_2 = a+b+c, \quad h_3 = a+b+e, \qquad a > 0, \quad b, c, e \geq 0. \tag{1}$$

Define degree-six polynomials

$$Q_6(x) = 6!F(x+2), \qquad Q_7(x) = \frac{7!F(x+2)}{2x+5}.$$

The quotient in dimension seven is a polynomial: binomial symmetry gives $F(-1-z) = (-1)^d F(z)$, so $F(-1/2) = 0$ when $d = 7$. In both cases write $Q_d(x) = \sum_{r=0}^{6} q_r x^{6-r}$. Expanding the binomial products gives the following exact coefficient matrices, whose rows are indexed by $r = 0, \dots, 6$:

$$(q_0, \dots, q_6)^T = C_d (a, b, c, e)^T, \tag{2}$$

$$C_6 = \begin{pmatrix} 7 & 5 & 2 & 1 \\ 105 & 75 & 30 & 15 \\ 1015 & 575 & 200 & 85 \\ 5775 & 2625 & 750 & 225 \\ 19138 & 6980 & 1598 & 274 \\ 34440 & 9900 & 1740 & 120 \\ 25920 & 5760 & 720 & 0 \end{pmatrix}, \qquad C_7 = \begin{pmatrix} 4 & 3 & 1 & 1 \\ 60 & 45 & 15 & 15 \\ 760 & 423 & 127 & 85 \\ 5100 & 2355 & 645 & 225 \\ 20836 & 7542 & 1744 & 274 \\ 46680 & 12960 & 2220 & 120 \\ 45360 & 9072 & 1008 & 0 \end{pmatrix}.$$

In particular $q_0, q_6 > 0$ throughout (1).

Excluding crossings of the imaginary axis

For a real sextic $Q(x) = q_0 x^6 + q_1 x^5 + \cdots + q_6$, define

$$D(Q) = \det \begin{pmatrix} q_1 & q_3 & q_5 & 0 & 0 \\ q_0 & q_2 & q_4 & q_6 & 0 \\ 0 & q_1 & q_3 & q_5 & 0 \\ 0 & q_0 & q_2 & q_4 & q_6 \\ 0 & 0 & q_1 & q_3 & q_5 \end{pmatrix}. \tag{3}$$

This is the fifth Hurwitz determinant, but the proof below needs no stability criterion as a black box.

Lemma 1. *If $q_6 \neq 0$ and $D(Q) \neq 0$, the polynomial Q has no root on the imaginary axis.*

Proof. Zero is not a root. If iy were a root for a nonzero real y, then $t = y^2$ would be a common root of

$$E(t) = -q_0 t^3 + q_2 t^2 - q_4 t + q_6, \qquad O(t) = q_1 t^2 - q_3 t + q_5.$$

Consequently the nonzero vector $(t^4, t^3, t^2, t, 1)^T$ would lie in the kernel of

$$K = \begin{pmatrix} -q_0 & q_2 & -q_4 & q_6 & 0 \\ 0 & -q_0 & q_2 & -q_4 & q_6 \\ q_1 & -q_3 & q_5 & 0 & 0 \\ 0 & q_1 & -q_3 & q_5 & 0 \\ 0 & 0 & q_1 & -q_3 & q_5 \end{pmatrix}.$$

Row permutations and column sign changes give $\det K = -D(Q)$, contradicting its nonzero value. □

For the two matrices (2), exact expansion of (3) has the form

$$D(Q_d) = g_d \sum_{i+j+\ell+m=5} A^{(d)}_{i,j,\ell,m} a^i b^j c^\ell e^m, \qquad g_6 = 3888000, \quad g_7 = 21168000. \tag{4}$$

The appendix gives every coefficient. Each is a positive integer; the smallest is 1176 for $d = 6$ and 216 for $d = 7$. There are exactly $\binom{8}{3} = 56$ monomials in each expansion. In particular the coefficient of a^5 is positive, so

$$D(Q_6) > 0, \qquad D(Q_7) > 0 \quad (a > 0,\ b, c, e \geq 0). \tag{5}$$

This is a finite exact algebraic certificate valid on the entire unbounded parameter domain, not extrapolation from numerical tests. It can be verified by substituting (2) into (3) and using the determinant's permutation formula. The accompanying `verify.py` does this using only integer arithmetic; it also reconstructs (2) directly from the binomial products and checks $\det K = -D(Q_d)$ independently.

A stable base polynomial and the full argument

At $b = c = e = 0$, all $h_j = a$. The polynomial binomial summation identity gives

$$a^{-1}F(z) = \sum_{j=0}^{d} \binom{z+d-j}{d} = \binom{z+d+1}{d+1} - \binom{z}{d+1}.$$

Every root of this difference has real part $-1/2$. Indeed, put $z = w - 1/2$ and $s_k = k + 1/2$, $0 \leq k \leq d$. The two products to be compared, up to the common factor $(d+1)!$, are $\prod_k (w + s_k)$ and $\prod_k (w - s_k)$. If $\operatorname{Re} w > 0$, then $|w + s_k| > |w - s_k|$ for every k, so the products cannot be equal. If $\operatorname{Re} w < 0$, all inequalities reverse. This proves the assertion, including the possibility that one product is zero. Thus all roots of Q_6 and Q_7 at this base point have real part $-5/2$; removal of the root $z = -1/2$ in the odd case does not affect this.

Fix any parameters (1) and replace (b, c, e) by (tb, tc, te), $0 \leq t \leq 1$. Along this continuous path, $q_0 > 0$ and $q_6 > 0$, and (5) and Lemma 1 exclude every imaginary-axis root. The degrees stay equal to six. After division by q_0, the coefficients vary continuously on a compact interval and remain bounded, so the roots have a uniform bound. Continuity of polynomial roots (equivalently, local invariance of the root count by the argument principle) implies that the number of roots in each open half-plane cannot change along the path. Initially all six have negative real part, so the same is true at $t = 1$.

Therefore every root of F has real part less than 2, including the removed root $-1/2$ when $d = 7$. The identity $F(-1 - z) = (-1)^d F(z)$ also sends every root z to $-1 - z$. Applying the upper bound to this reflected root gives $\operatorname{Re} z > -3$. This proves Theorem 1 and the requested strict inequality $|\operatorname{Re}(z) + 1/2| < 5/2$. □

Four proposed follow-up tasks

Task 2. Determine the sharp supremum of $|\operatorname{Re} z + 1/2|$ among roots with both nonzero imaginary part and $\operatorname{Re} z \neq -1/2$, separately for the two coefficient cones in Theorem 1.

Task 3. Decide the half-strip assertion in dimensions eight and nine, first for palindromic coefficient vectors satisfying Hibi's lower bounds, and then for vectors realized by reflexive lattice polytopes.

Task 4. Characterize which boundary coefficient directions that approach the extremal strip in dimensions six and seven can be realized by reflexive polytopes, and obtain any resulting sharper geometric bound.

Task 5. Find an analytic factorization or positivity representation for the determinant in (3) that explains the two positive coefficient expansions and identifies the dimensions in which it persists.

References

[1] G. Hegedűs, A. Higashitani, A. M. Kasprzyk, *Ehrhart polynomial roots of reflexive polytopes*, arXiv:1503.05739v1 (2015), Conjecture 6.4, p. 15.

[2] G. Hegedűs, A. Higashitani, A. M. Kasprzyk, *Ehrhart polynomial roots of reflexive polytopes*, Electronic Journal of Combinatorics 26(1) (2019), P1.38, doi:10.37236/7780.

Complete coefficient certificate for the proof

The table gives the coefficients of Δ_5/g_5 in the monomials $a^i b^j c^\ell e^m$. The entries in both columns are strictly positive. All $\binom{8}{3} = 56$ quadruples of nonnegative integers of total degree five are included; there are no omitted terms.

Order $k = 5$

The contents are $g_{6,5} = 3888000$ and $g_{7,5} = 21168000$.

i	j	ℓ	m	Dimension 6	Dimension 7
5	0	0	0	12007299472	1014099264
4	1	0	0	28646940070	2424283200
4	0	1	0	8630859020	673254720
4	0	0	1	2920602860	300987360
3	2	0	0	27146887075	2308495948
3	1	1	0	16198103355	1278257064
3	1	0	1	5355975940	544060320
3	0	2	0	2382863560	176566428
3	0	1	1	1522428460	145995120
3	0	0	2	221387740	13393620
2	3	0	0	12836637375	1103128788
2	2	1	0	11433519350	917095440
2	2	0	1	3729969800	386039236
2	1	2	0	3363248020	254116836
2	1	1	1	2138889745	212353908
2	1	0	2	315274930	27039540
2	0	3	0	327965405	23492376
2	0	2	1	306772645	29568816
2	0	1	2	89293010	8627220
2	0	0	3	8005660	416340
1	4	0	0	3034762500	264730950
1	3	1	0	3595816625	293721621
1	3	0	1	1164641125	124492464
1	2	2	0	1587398100	122163276
1	2	1	1	1008550225	103728485
1	2	0	2	150667775	15868184
1	1	3	0	310189380	22567365
1	1	2	1	291197545	28917138
1	1	1	2	86188385	9583767
1	1	0	3	7999260	761160
1	0	4	0	22670725	1560708
1	0	3	1	28041725	2683953
1	0	2	2	12359950	1410534
1	0	1	3	2287300	259695
1	0	0	4	147700	9450
0	5	0	0	287100000	25527150
0	4	1	0	424762500	35413605
0	4	0	1	137137500	15259320
0	3	2	0	250160500	19629588
0	3	1	1	159141750	17006589
0	3	0	2	24080750	2950548
0	2	3	0	73422075	5430589

i	j	ℓ	m	Dimension 6	Dimension 7
0	2	2	1	69242300	7102879
0	2	1	2	20807775	2593925
0	2	0	3	1988650	267632
0	1	4	0	10748395	749148
0	1	3	1	13388115	1313760
0	1	2	2	6001955	746376
0	1	1	3	1145535	166011
0	1	0	4	78120	12030
0	0	5	0	628082	41184
0	0	4	1	970305	90612
0	0	3	2	577430	70371
0	0	2	3	165165	24597
0	0	1	4	22610	3870
0	0	0	5	1176	216